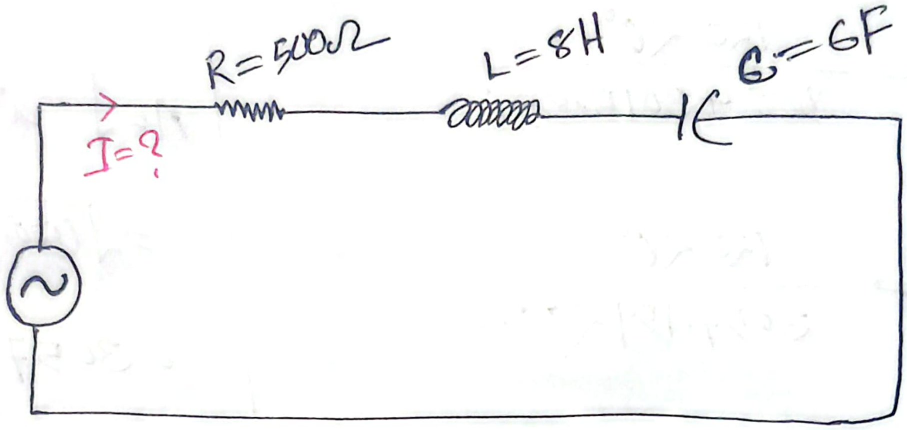
AC



$$V = \underbrace{120}_{V_m} \cos(\underbrace{377}_{\omega} t + \underbrace{0^\circ}_{\phi})$$

we know that,

$$\cancel{V = IZ}$$

$$Z = R + (X_L - X_C)i$$

$$= 500 + (3016 - 0.0004)i$$

$$= 500 + 3015.99i$$

$$X_L = \omega L$$

$$= 377 \times 8$$

$$= 3016\ \Omega$$

$$X_C = \frac{1}{\omega C}$$

$$= \frac{1}{377 \times 6}$$

$$= 0.0004\ \Omega$$

we know that,

$$V = IZ$$

$$\Rightarrow I = \frac{V}{Z}$$

$$= \frac{120\cos(377t + 0^\circ)}{500 + 3015.99i}$$

$$= \frac{120 \angle 0^\circ}{500 + 3015.99i}$$

$$= \frac{120 \angle 0^\circ}{3057.154 \angle 80.59^\circ}$$

$$= \frac{120}{3057.154} \angle 0^\circ - 80.59^\circ$$

$$= 0.039 \angle -80.59^\circ$$

$$A = \sqrt{a^2 + b2}$$

$$= \sqrt{(500)^2 + (3015.99)^2}$$

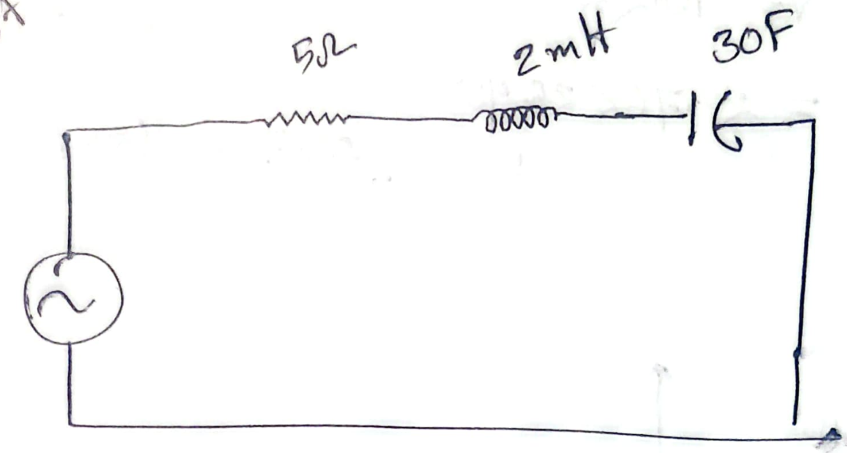
$$= 3057.154$$

$$\theta = \tan^{-1}\left(\frac{3015.99}{500}\right)$$

$$= 80.59^\circ$$

$v = V_m \cos(\omega t + \phi)$

$v = 120 \cos(200t + 30^\circ)$



$V = IZ$

$I = \frac{V}{Z}$

$V = V_m \angle \phi$

$= 120 \angle 30^\circ$

$X_L = \omega L$

$X_C = \frac{1}{\omega C}$

$$Z = R + i(X_L - X_C)$$

$$= 5 + i\left(200 \times 2 \times 10^{-3} - \frac{1}{200 \times 30}\right)$$

$$\boxed{Z = 5 + 0.4i}$$

$$I = \frac{V}{Z}$$

$$= \frac{120 \angle 30^\circ}{5 + 0.4i}$$

a b

$$= \frac{120 \angle 30^\circ}{\sqrt{5^2 + 0.4^2} \angle \tan^{-1}\left(\frac{0.4}{5}\right)}$$

$$= \frac{120 \angle 30^\circ}{5.015 \angle 4.58^\circ}$$

$$\boxed{I = 2376 \angle 25.43^\circ}$$

$X_L = X_C$

$$f_R = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi \times \sqrt{2 \times 10^{-3} \times 30}}$$

$= 0.65\ Hz$

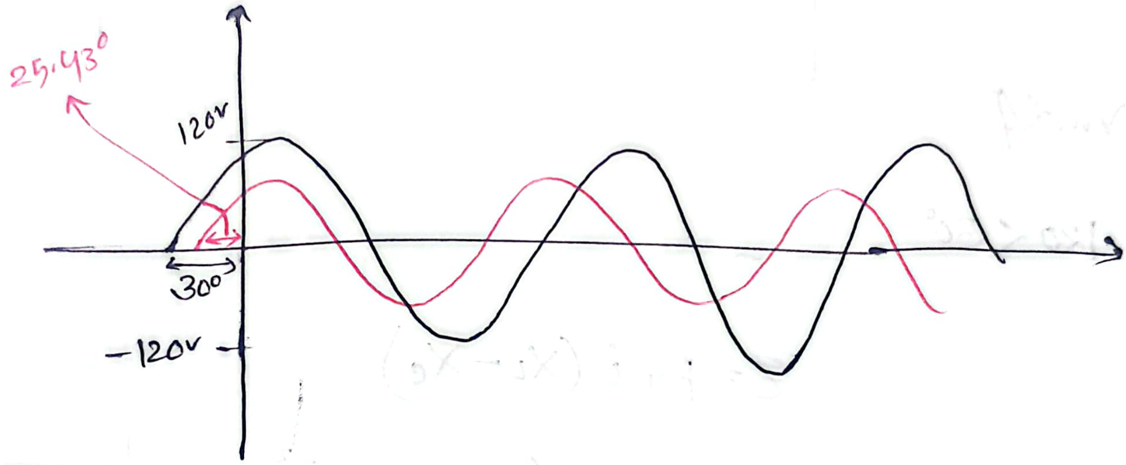
$\omega = 2\pi f$

$200 = 2\pi f$

$f = \frac{200}{2\pi} = 31.83\ Hz$

$V = 120\angle 30^\circ \rightarrow 120\cos(200t + 30^\circ)$

$I = 23.76\angle 25.43^\circ \rightarrow 23.76\cos(200t + 25.40^\circ$



$V = 120 \cos(200t + 30^\circ)$

$I = 23.9 \cos(200t + 25.43^\circ)$

$$30^\circ - 25.43^\circ = 4.57^\circ$$

$\therefore$ V leads I by 4.57°

I lags V by 4.57°